

DELIGHT Reconstruction ML

Sebastian Sjoström and Leon Windett

Overview

- Experimented with implementing different approaches to cross-channel and intra-channel processing.
- Trained model on 2D position reconstruction, energy regression and recoil classification (on both small and large dataset).
- Conducted full evaluation of encoder-only based model (on small dataset).

- Github: https://github.com/sebspel/DELIGHT_reconstruction.git

Resolving Normalisation Challenges

- Attempted normalising both energy and position targets.
- Tested normalising energy with $\log_{10}$ and z-score.
- Similarly, tested normalising position with z-score.
- Training instability issues resolved by removing all normalisation.

Temporal Processing

- Encoder-only model uses Rotary Positional Embedding (RoPE) (Su et al., 2023) based encoder blocks for intra channel feature extraction.
- Also trained both unidirectional and bidirectional selective state space blocks (SSMs) with Mamba 2 (Dao and Gu, 2024) to leverage its inductive bias.
- Both encoder-only and Mamba 2-encoder hybrids have patches compressed into a learned query vector via attention pooling.

Mamba/Mamba 2 Blocks

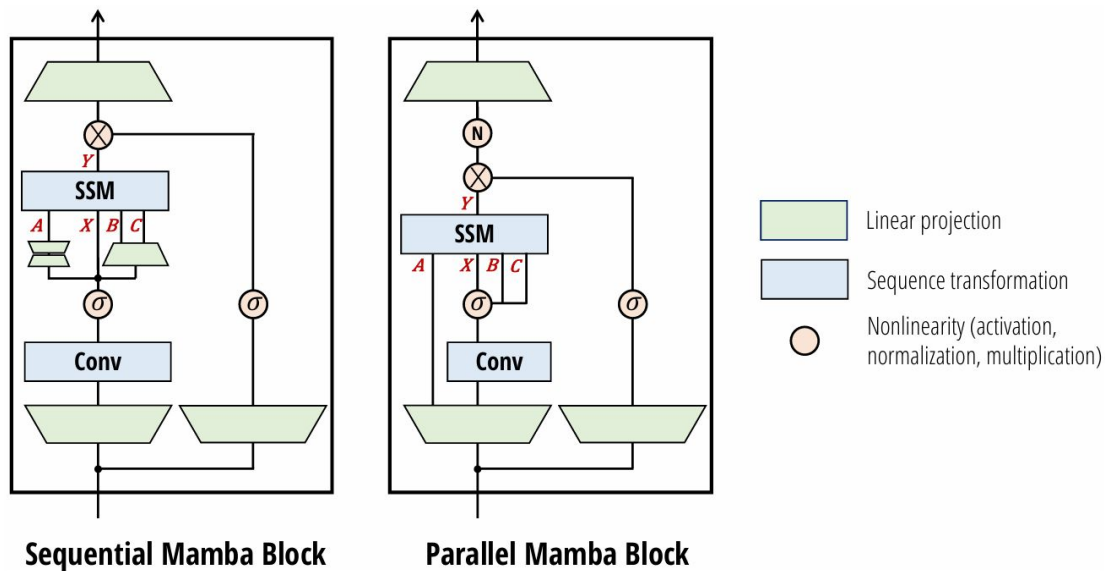


Figure from Dao and Gu (2024)

Spatial Processing

- Encoder-only model processed channels as sequence tokens, learning attention bias from pairwise position differences of detectors.
- Trained graph neural network (GNN) blocks where each channel is an explicit graph node to leverage physical inductive bias.
- Encoder-GNN hybrid introduces richer spatial encoding (10 edge features).

Preliminary Assessment

- Encoder-only, encoder-GNN and SSM-encoder models perform well on all three tasks.
- Encoder-only runs significantly faster and is easier to scale upwards.
- On balance, encoder-only outperforms hybrid models on small dataset.

Encoder-only Results (large validation set)

	2D Spatial RMSE (mm)	Energy RMSE (eV)	Recoil Accuracy
Previous model (large dataset)	8.82	127	N/A
Current model (large dataset)	6.31	17.2	91.2%

Encoder-only Results (small test set)

- Model performs well for 2D position reconstruction on all energy scales, (14.8 mm RMSE at 20 eV).
- Energy regression and classification performance drops off rapidly at 20 eV energy scales e.g. ROC-AUC of 0.568 at 20 eV vs 0.964 at 50 eV.
- Minimum energy resolution of 0.118 at 500 eV as expected (highest SNR).

Next Steps

- Train and evaluate SSM-encoder hybrid on large dataset.
- Iterate on existing encoder-only model.
- Experiment with applying denoising techniques to sensor data then training and evaluating models on the processed data.

References

J. Su, M. Ahmed, Y. Lu, S. Pan, W. Bo and Y. Liu, RoFormer: Enhanced transformer with Rotary Position Embedding, Neurocomputing 568, 127063 (2024), doi:10.1016/j.neucom.2023.127063.

T. Dao and A. Gu: Transformers are SSMs: Generalized Models and Efficient Algorithms Through Structured State Space Duality, Proc. ICML (2024), doi:10.48550/arXiv.2405.21060.